

LAB EXPERIMENT RECORD

SET 2: CUSTOMER FEEDBACK ANALYSIS

Department	Artificial Intelligence and Data Science
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 2: CUSTOMER FEEDBACK ANALYSIS

Dataset: Customer Satisfaction

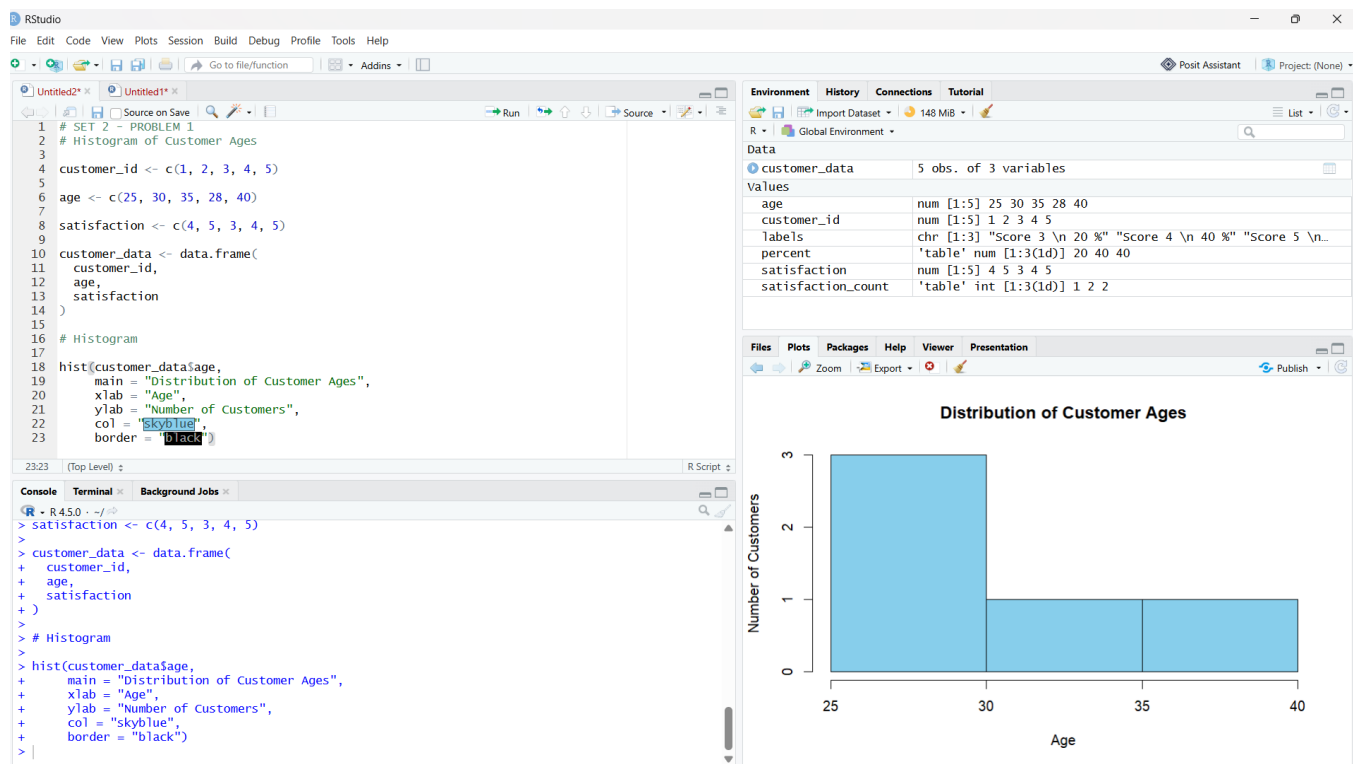
Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# SET 2 - PROBLEM 1
# Histogram of Customer Ages
customer_id <- c(1, 2, 3, 4, 5)
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)
customer_data <- data.frame(
  customer_id,
  age,
  satisfaction
)
# Histogram
hist(customer_data$age,
  main = "Distribution of Customer Ages",
  xlab = "Age",
  ylab = "Number of Customers",
  col = "skyblue",
  border = "black")
```



Output 1: Distribution of Customer Ages

Observation: The histogram represents the distribution of the five customer ages in the given dataset.

PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores and include labels.

```
# Set 2 - Problem 2
# Customer Satisfaction Pie Chart

customer_id <- c(1, 2, 3, 4, 5)

age <- c(25, 30, 35, 28, 40)

satisfaction <- c(4, 5, 3, 4, 5)

customer_data <- data.frame(
  customer_id,
  age,
  satisfaction
)

print(customer_data)

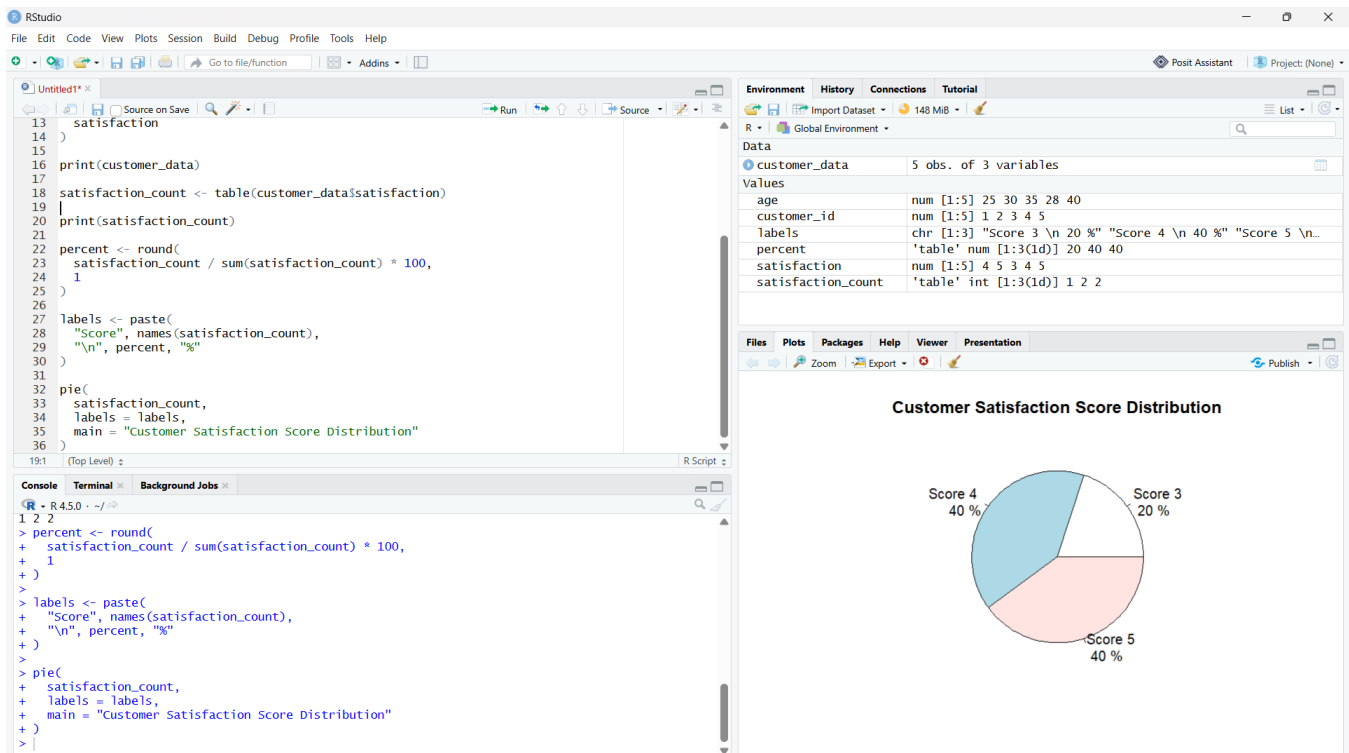
satisfaction_count <- table(customer_data$satisfaction)

print(satisfaction_count)

percent <- round(
  satisfaction_count / sum(satisfaction_count) * 100,
  1
)

labels <- paste(
  "Score", names(satisfaction_count),
  "\n", percent, "%"
)

pie(
  satisfaction_count,
  labels = labels,
  main = "Customer Satisfaction Score Distribution")
```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: Scores 4 and 5 each occur twice, while score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
# SET 2 - PROBLEM 3

# Stacked Bar Chart of Satisfaction by Age Group

customer_id <- c(1, 2, 3, 4, 5)

age <- c(25, 30, 35, 28, 40)

satisfaction <- c(4, 5, 3, 4, 5)

customer_data <- data.frame(
  customer_id,
  age,
  satisfaction
)

# Create age groups

customer_data$age_group <- cut(
  customer_data$age,
  breaks = c(20, 29, 39, 49),
  labels = c("20-29", "30-39", "40-49")
)

# Display data

print(customer_data)

# Stacked bar chart

ggplot(customer_data,
  aes(x = age_group,
    fill = factor(satisfaction))) +

  geom_bar() +

  labs(
```

```

title = "Customer Satisfaction Scores by Age Group",

x = "Age Group",

y = "Number of Customers",

fill = "Satisfaction Score"

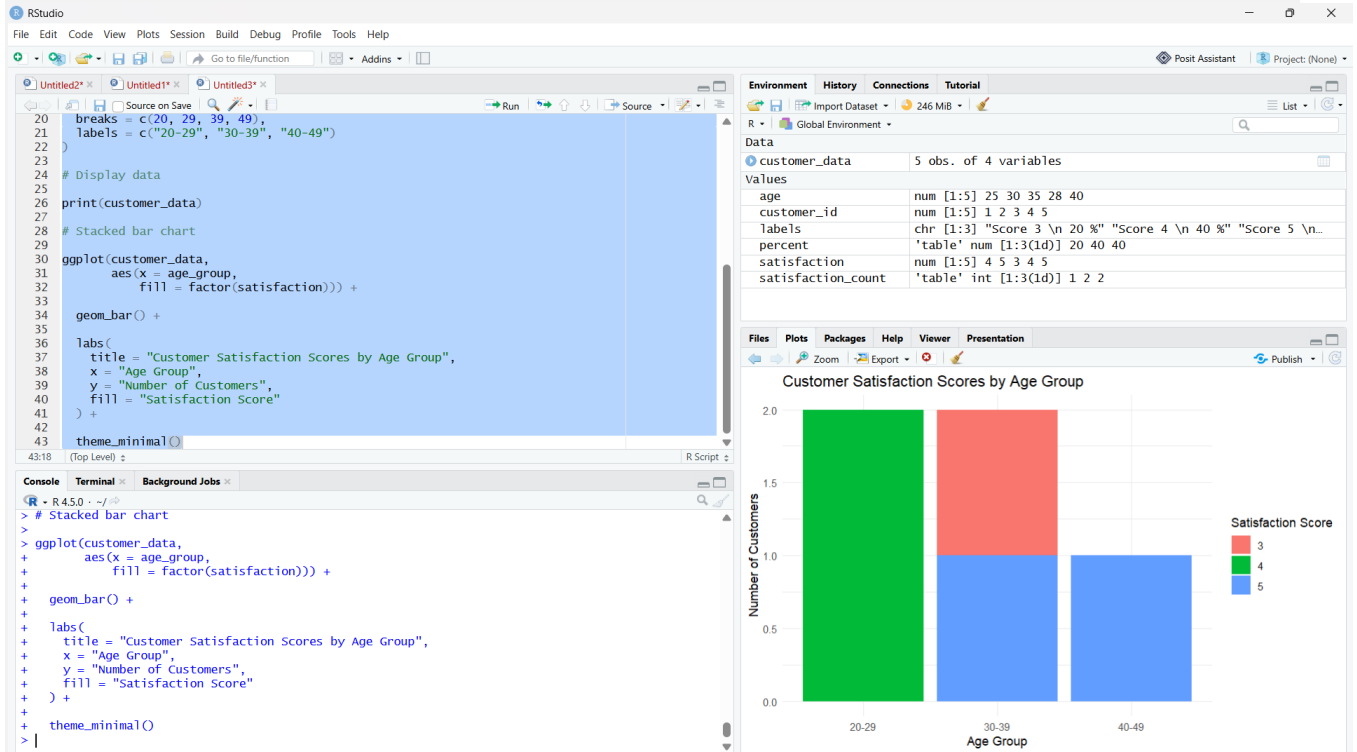
) +

```

```

theme_minimal()

```



Output 3: Satisfaction Scores by Age Group

Observation: The code creates three age groups and stacks the satisfaction-score frequencies for each group.

Note: The supplied RStudio screenshot is inserted exactly as captured, including its displayed x-axis labels.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
# Combine all feedback

text <- paste(feedback, collapse = " ")

# Convert to lowercase

text <- tolower(text)

# Remove punctuation

text <- gsub("[[:punct:]]", " ", text)

# Split into individual words

words <- unlist(strsplit(text, "\\s+"))

# Remove common words

stop_words <- c(
  "the", "and", "was", "were", "with",
  "very", "a", "an", "is", "i", "am",
  "to", "but", "my", "of", "for"
)

words <- words[!words %in% stop_words]

# Count words

word_frequency <- table(words)

# Create word cloud

wordcloud(
  words = names(word_frequency),
  freq = as.numeric(word_frequency),
  min.freq = 1,
```

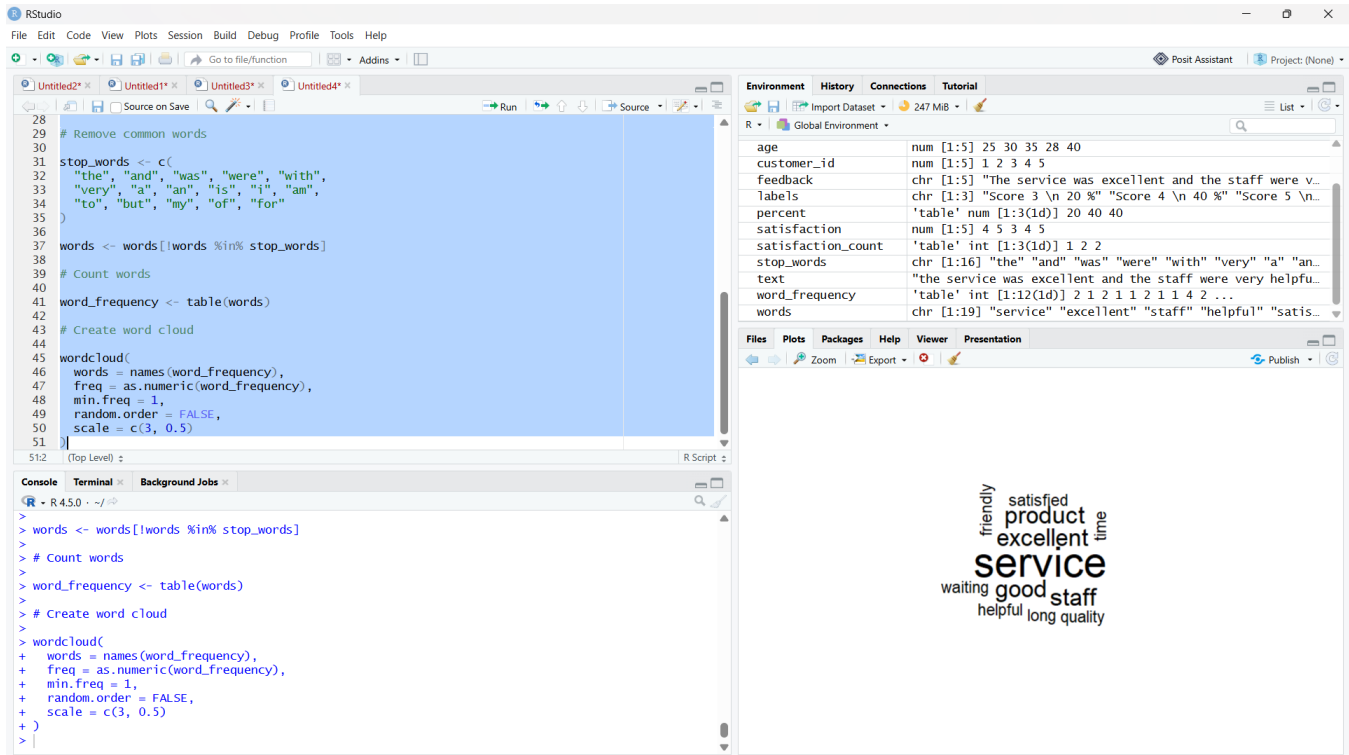
```

random.order = FALSE,

scale = c(3, 0.5)

)

```



Output 4: Customer Feedback Word Cloud

Observation: The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Set 2 Procedure

1. Create vectors for customer age and satisfaction score.
2. Use `hist()` to represent the distribution of customer ages.
3. Use `table()` and `pie()` to display the satisfaction score distribution.
4. Create age groups using `cut()`, form a frequency table, and use `barplot()` for the stacked bar chart.
5. Prepare open-ended feedback text, count the words, and use `wordcloud()` to visualize frequent terms.

Set 2 Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, stacked bar chart, and word cloud.